Md Shazid Islam

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Summary

Ph.D. candidate building vision-language and large language model systems for scientific data. I fine-tune VLMs and LLMs with preference optimization, design LLM-driven agentic pipelines that call executable tools, and ground generated outputs in retrieved evidence, alongside data-efficient imaging work in segmentation, domain adaptation, and active learning. First-author publications at MICCAI and in IEEE Transactions. I carry problems from formulation and algorithm design through rigorous experimentation, ablation, multi-GPU distributed training, and open-source release on GitHub.

Education

Ph.D. in Electrical and Computer Engineering (CGPA: 3.81) — September 2020 – Present
University of California, Riverside
UCR Vision and Learning Group
Advisor: Amit K. Roy-Chowdhury

M.Sc. in Electrical and Electronic Engineering (CGPA: 4.00) — October 2017 – July 2020
Bangladesh University of Engineering and Technology (BUET)
Advisor: S.M. Mahbubur Rahman

B.Sc. in Electrical and Electronic Engineering (CGPA: 3.89) — February 2013 – September 2017
Bangladesh University of Engineering and Technology (BUET)
Advisor: Mohammad Ariful Haque

Technical Skills

- **LLMs & Vision-Language Models:** Fine-Tuning (SFT, LoRA-style adaptation), Preference Optimization (DPO, GRPO), Instruction Tuning, Multimodal Fusion, Grounded / Image-Conditioned Generation; Qwen2.5, Qwen3, LLaVA, CLIP, SAM
- **Agentic Systems & Retrieval:** ReAct agent framework, tool use, LLM-driven executable-program pipelines, Retrieval Augmented Generation (RAG)
- **Machine Learning Methods:** Deep Learning, Transformers, CNNs, GNNs, GANs, Generative Models, Transfer Learning, Domain Adaptation, Active Learning
- **Mathematical Foundations:** Statistical Inference, Convex Optimization, Stochastic Processes, Estimation Theory, Information Theory
- **Languages:** Python, C/C++, MATLAB
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, OpenCV, NumPy, Pandas
- **Training & Infrastructure:** Multi-GPU parallel (distributed) training, CUDA, HPC, Docker, Git, TensorBoard
- **Application Domains:** Computer Vision, NLP, Multimodal Learning, Generative Modeling, Distributed / Federated Learning

Research Experience

Language-Guided Active Learning for Medical Image Segmentation (LINGUAL) — March 2025 – December 2025
- Developed an LLM-driven annotation pipeline that replaces pixel-wise labeling with natural-language feedback from clinical/domain experts.
- Used an LLM to parse expert instructions and invoke executable refinement tools (novel EXPAND and SHRINK modules) that apply the requested label corrections.
- Reduced annotation interaction cost by up to approximately 25.27% while maintaining or improving segmentation performance over manual baselines, under domain shift and limited supervision.

Uncertainty Quantification for 3D Vision-Language Models — August 2026 – Present
- Developing a training-free uncertainty quantification method for 3D medical vision-language models that attributes each word of a generated CT report to the volumetric subregions that support or contradict it, using block-wise feature replacement (occlusion) on a 3D ViT token grid.
- Designed an evidential framework that pools per-region evidence into word-level uncertainty (vacuity, dissonance) and aggregates it into a scan-level uncertainty map, producing a budgeted 3D review region that directs radiologist attention to the most conflicting anatomy.
- Built the full pipeline on large-scale CT data (NV-Reason-CT on AbdomenAtlas 3.0, 900+ volumes on a SLURM GPU cluster), including NIfTI overlay visualization in 3D Slicer and a label-free evaluation protocol with clinically grounded error labels.

Vision-Language Models for Automated Radiology Report Generation — January 2026 – May 2026
- Built an end-to-end training and evaluation pipeline that fine-tunes vision-language models (LLaVA-Med 1.5) for grounded, image-conditioned report generation, trained on MIMIC-CXR with multi-GPU parallel (distributed) training, using supervised fine-tuning and preference optimization (DPO, GRPO).
- Enforced image-grounded reasoning to reduce hallucinated findings and improve spatial factuality over raw-image baselines, which matters where outputs must be trustworthy.

Online Domain Adaptation for Medical Image Segmentation (ODES) — July 2023 – June 2024 [Code]
- Proposed a source-free, single-pass online domain adaptation framework that corrects unreliable pseudo-labels with minimal expert

guidance and no additional storage, addressing scanner/site distribution shift common in clinical data.
- Designed a novel image pruning strategy to select which images require annotation and a spatial and feature-wise diversity-aware acquisition mechanism to maximize annotation efficiency.
- Demonstrated consistent gains over test-time adaptation (+7.7% on average), comparable to active domain adaptation across multiple medical datasets.

3D Biomedical Image Analysis from Microscopy (DEGAST3D) June 2021 – June 2022 [Code]
- Designed a deformable 3D graph neural network to track densely packed plant cells in unregistered 3D time-lapse microscopy, improving precision +6.83%, recall +5.96%, and F1 +6.40% over state-of-the-art.
- Built a 3D registration and cell division detection pipeline, improving recall +15.38% and F1 +14.78% over 2D approaches on a public benchmark.
- Performed 3D segmentation and 3D reconstruction of volumetric plant cells.

Additional Research (Breadth Across Domains)
- **Generative & Multimodal Modeling** (Jan 2019 – Jul 2020): Developed a CNN-LSTM-Mixture Density Network generating rhythm-aligned motion from audio, rendered to video via a Pix2pixHD GAN; outperformed prior methods across four motion styles.
- **Multi-Human Tracking** (Jul 2022 – Jun 2023): Built a multi-person tracking pipeline using ByteTrack for detection-based association, with person re-identification and gait analysis to maintain identities across frames.
- **Distributed / Federated Learning** (Jun 2024 – Feb 2025): Built a delay-aware asynchronous federated training framework that mitigates stragglers via buffer diversification and contribution-based weighting on decentralized data.

Selected Publications

- **Md Shazid Islam**, Shreyangshu Bera, Sudipta Paul, Amit K. Roy Chowdhury. "LINGUAL: Language-INtegrated GUidance in Active Learning for Medical Image Segmentation." *WACV 2027.* [Preprint]
- **Md Shazid Islam**, Sayak Nag, Arindam Dutta, Miraj Ahmed, Fahim Faisal Niloy, Shreyangshu Bera, Amit K. Roy Chowdhury. "ODES: Online Domain Adaptation with Expert Guidance for Medical Image Segmentation." *MICCAI 2025.* [Paper] [Code]
- **Md Shazid Islam**, SM Mahbubur Rahman. "Synthesis of Dance by Learning Body Gestures from Music." *Multimedia Tools and Applications, September 2025.* (Q1) [Paper]
- **Md Shazid Islam**, Arindam Dutta, Calvin-Khang Ta, Kevin Rodriguez, Christian Michael, Mark Alber, G. Venugopala Reddy, Amit K. Roy Chowdhury. "DEGAST3D: Learning Deformable 3D Graph Similarity to Track Plant Cells in Unregistered Time Lapse Images." *IEEE Trans. Computational Biology and Bioinformatics, 2025.* (Q1) [Paper] [Code]
- Md Saiful Islam Sajol, ASM Jahid Hasan, **Md Shazid Islam**, Md Saydur Rahman. "A ConvNeXt V2 Approach to Document Image Analysis: Enhancing High-Accuracy Classification." *IEEE CITDS, Debrecen, Hungary, 2024.* [Paper]
- Md Saiful Islam Sajol, ASM Jahid Hasan, **Md Shazid Islam**, Md Saydur Rahman. "Transforming Social Media Analysis: TweetEval Benchmarking with Advanced Transformer Models." *ISMSIT, Ankara, Turkey, 2024.* [Paper]
- Md Sanzid Bin Hossain, **Md Shazid Islam**, Md Saad Ul Haque, Md Saydur Rahman. "Gait Phase Classification from sEMG in Multiple Locomotion Mode Using Deep Learning." *9th Int. Congress on Information and Communication Technology, London, 2024.* [Paper]
- **Md Shazid Islam**, ASM Jahid Hasan, Md Saydur Rahman, Jubair Yusuf, Md Saiful Islam Sajal, Farhana Akter Tumpa. "Location Agnostic Source-Free Domain Adaptive Learning to Predict Solar Power Generation." *IEEE ETFG, Wollongong, Australia, 2023.* [Paper]
- Shreyangshu Bera, **Md Shazid Islam**, Saketh Bachu, B. S. Manjunath, Amit Roy-Chowdhury. "SALT: Spectral Instability based Active Learning for Tiny Object Detection." *Under Review.* [Preprint]
- **Md Shazid Islam**, M Yashwanth, Xiangyu Chang, Anirban Chakraborty, Srikanth Krishnamurthy, Amit Roy-Chowdhury. "DAAFL: Delay Aware Asynchronous Federated Learning." *Under Review.*

Leadership

- **Data Science Challenge, Lawrence Livermore National Laboratory (LLNL)** (2026): Led a four-member team (Team Agent Lattice) building an LLM-driven agentic workflow on defect detection in materials. Directed task decomposition and integration across teammates .

Honors and Awards

- MICCAI NIH Registration Grant, 2025
- Dean's Distinguished Fellowship, University of California, Riverside
- Dean's List Scholarship, Department of Electrical and Electronic Engineering, BUET
- Merit Scholarship, Department of Electrical and Electronic Engineering, BUET

Graduate Courses

Deep Learning, Advanced Computer Vision, Design and Analysis of Algorithms, Convex Optimization, Pattern Recognition, Stochastic Processes, State and Parameter Estimation Theory, Information Theory, Math Methods for Engineering

References

- **Amit K. Roy-Chowdhury** – Professor, ECE; Co-Director, RAISE Institute, UC Riverside. vcg.ece.ucr.edu/amit
- **B. S. Manjunath** – Distinguished Professor, ECE, UC Santa Barbara. engineering.ucsb.edu/people/bs-manjunath
- **Venugopala Reddy Gonehal** – Professor, Botany and Plant Sciences, UC Riverside. profiles.ucr.edu/venug